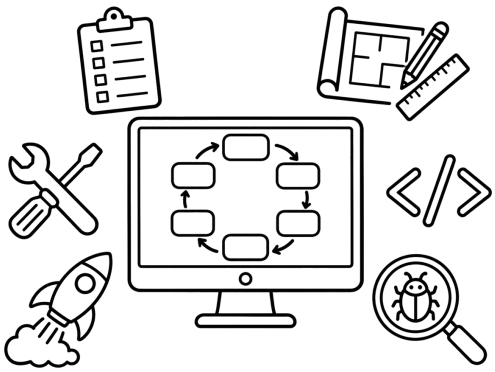
ICT

INFORMATION AND COMMUNICATION TECHNOLOGY

GRADE 11



UNIT 2
System Development Life Cycle

Student Workbook

"The King of ICT"

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STUDENT DETAILS

Name: _____

School: _____

Institution: _____

Contact: _____

BEFORE WE START

What This Unit Is About

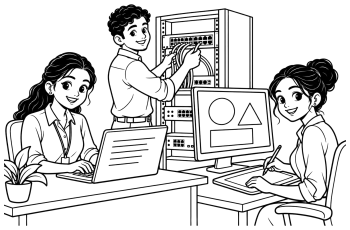
Read this page with your teacher

AT THE END OF THIS UNIT YOU WILL BE ABLE TO

- [ ] Explain what a system is and name its three basic components.
- [ ] Find the input, the process and the output of a system.
- [ ] Explain information systems: manual and computer based.
- [ ] Compare manual systems with computer based information systems.
- [ ] Name the six steps of the System Development Life Cycle (SDLC).
- [ ] Explain how a new system is planned, tested, deployed and maintained.
- [ ] Compare the waterfall model and the iterative incremental model.

HOW TO USE THIS BOOK

Read each chapter with your teacher, do the practical tasks and then write the exercises. The **WRITING SPACES** are for the notes your teacher gives you in class - write them neatly: they are your revision notes for the O/L examination.



THE WORDS YOU WILL LEARN

System	Component	Input	Process
Output	Data	Information	Manual system
Computer based	SDLC	Requirements	System analyst
Observation	Interview	Questionnaire	Prototype
Debugging	Unit testing	Deployment	Pilot
Phased	Maintenance	Waterfall model	Iterative

"A good system is like a good team: every part does its job, and all the parts work together."

- THE KING OF ICT

"Neat notes today are your best friend in the O/L examination."

- THE KING OF ICT

CHAPTER 1

Systems and Information Systems

? THE LIBRARY QUEUE

"We wait so long at the library counter. Every book is written in a big register!"

"A computer based library system would find a book and its place in seconds."

A library, a school and a weather station are all **systems**. Meet systems and information systems.



2.1 What is a system?

A **system** is a collection of **components** that interact to achieve a specific task. In Figure 2.1 the parts of the road construction machine are not interconnected, so the machine does not function and road construction is not possible. In Figure 2.2 the parts are interconnected: the machine works as a **system** and achieves its goals.

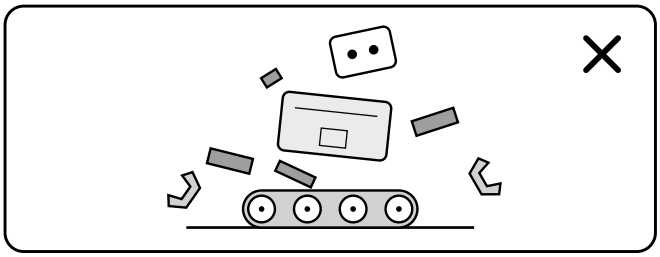


Figure 2.1 - the parts are not interconnected
It does not work.

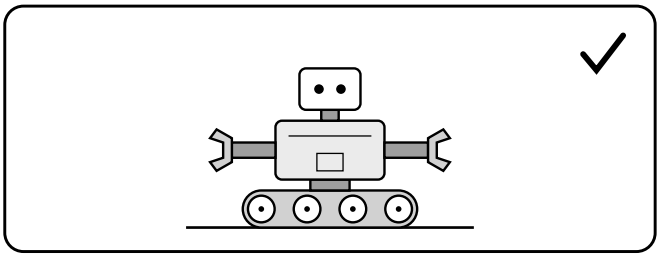


Figure 2.2 - the parts are interconnected
It works as a system.

2.2 The basic elements of a system

A system consists of three basic components: **input**, **process** and **output**. The input received by a system is converted to output by processing.



Figure 2.3 - the basic components of a system

Example 1 - the school as a system. Its objective: to produce a responsible, worthwhile, just citizen to society. **Example 2** - a weather forecasting system outputs tomorrow's weather:

	Example 1 - the school	Example 2 - weather forecasting
Input	Children	Atmospheric pressure, temperature, wind direction, humidity
Process	The child is subjected to the teaching/learning process through interactions of teachers and other resources	The weather forecasting system
Output	Providing good citizens to the country	Tomorrow's weather

WRITING SPACE 1 - What a system is write what your teacher tells you

2.3 Information systems

A system which converts **data** into **information** is known as an **information system**.

Information systems can be classified as **manual information systems** and **computer based information systems**.



Figure 2.4 - an information system

2.4 Manual information systems

In a **manual system** all processes are done manually. Example: in a manual student information system, each child has a registered number and a personal file (personal information, special abilities, achievements). The principal requires a progress report on a student for the previous year:

INPUT	PROCESS	OUTPUT
Name of student and relevant year	1. Find the index number of the student. 2. Find the personal file according to the index number. 3. Extract the required information for the relevant year from the file. 4. Prepare a report using this information.	Progress report on the student for the relevant year

2.5 Computer based information systems

A system which converts data into information **using a computer** is known as a **computer based information system**. Consider a computer based library information system. Some of its benefits to a librarian:

- It detects whether a particular book is available in the library and, if it is, where the book is.
- A bar code is used to maintain issues and receipts.
- Fines for late returns are calculated automatically.
- Books can be searched for efficiently.
- Records of those who have borrowed books are maintained.
- Those who have failed to return books on the due dates can be listed.
- An online system for remote reservation and searching for books.
- E-books can be provided to members.

2.6 Manual or computer based?

The main differences between manual systems and computer based information systems:

Computer based information systems	Manual systems
Errors are minimal with data processed by a computer program.	Since the data is manually processed, there is more room for error.
Data can be processed more efficiently.	Processing of data is less efficient.
Large amounts of data can be stored in a small physical space. Database software can be used for the purpose.	Requires a large space for data storage. Filing cabinets are required to store files.
Security can be ensured with backups and the use of passwords.	Data is open to a lot of threats. It is not as safe as a computer based system.

WRITING SPACE 2 - Manual or computer based? write what your teacher tells you

DO IT - PRACTICAL 1 (TEXTBOOK ACTIVITIES, PAGES 55 AND 56)

Manual or computer based? Work in pairs, with your teacher.

1	Read the manual student information system of 2.4 again: its input, process and output.
2	Imagine the school uses a computer based information system instead. Write the advantages and disadvantages of getting the progress report from each system in the table below (use 2.6).
3	Visit the school library. List the benefits a computer based library information system would give the librarian (2.5).
4	Compare and contrast your answers with the class.

System	Advantages	Disadvantages
Manual system		
Computer based system		

"Data becomes useful only when a good system turns it into information."

- THE KING OF ICT

A Input, process or output?

8 marks

Look at the examples of 2.2 and 2.4. Tick one box for each line.

It is the ...	Input	Process	Output
Children (the school system)	☐	☐	☐
Providing good citizens to the country (the school system)	☐	☐	☐
The teaching/learning process with teachers and other resources	☐	☐	☐
Atmospheric pressure, temperature, wind direction and humidity	☐	☐	☐
Tomorrow's weather (the weather forecasting system)	☐	☐	☐
Name of student and relevant year (the student information system)	☐	☐	☐
Find the personal file according to the index number	☐	☐	☐
Progress report on the student for the relevant year	☐	☐	☐

B Put the process steps in order

4 marks

The manual student information system: write 1, 2, 3 and 4 in the boxes.

a. ☐ Prepare a report using this information.

b. ☐ Find the index number of the student.

c. ☐ Extract the required information for the relevant year from the file.

d. ☐ Find the personal file according to the index number.

C Manual or computer based?

8 marks

Which system is it? Tick one box for each line.

The system ...	Manual	Computer based
Filing cabinets are required to store the files.	☐	☐
Security can be ensured with backups and passwords.	☐	☐
Since the data is processed by hand, there is more room for error.	☐	☐
Large amounts of data can be stored in a small physical space.	☐	☐
Data is open to a lot of threats.	☐	☐
Fines for late returns are calculated automatically.	☐	☐
All the processes are done manually.	☐	☐
Data can be processed more efficiently.	☐	☐

"One part alone cannot do the work of the whole system."

- THE KING OF ICT

D Choose the correct answer

5 marks

Underline your answer.

1. The three basic components of a system are
   (a) data, information, report (b) input, process, output
   (c) files, cabinets, reports (d) input, output, storage

2. In the weather forecasting system, humidity is
   (a) an input (b) a process (c) an output (d) an objective

3. Information systems are classified as manual and
   (a) paper based (b) computer based (c) school based (d) file based

4. Which is NOT a benefit of a computer based library system?
   (a) calculating fines (b) searching for books
   (c) e-books for members (d) more filing cabinets

5. Which is true for a computer based information system?
   (a) open to many threats (b) needs a large space
   (c) efficient processing (d) more room for error

CHAPTER 2

The System Development Life Cycle

? WHERE DO WE START?

"The principal wants a computer based library system. Where do we start?"

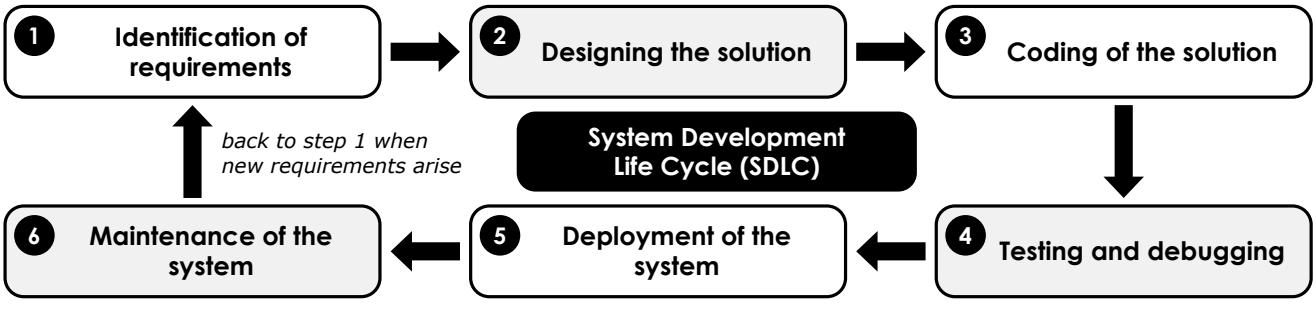
"First we find out what the librarian needs. Let us meet her for a discussion."

A system is developed step by step: the **System Development Life Cycle**.



2.7 The system development life cycle

To develop an information system there are different methodologies. The **System Development Life Cycle** (SDLC) is one aspect of the process. It has six steps, and it can be **iterated**: the process goes back to the first step when new requirements arise.



WRITING SPACE 3 - The six steps of the SDLC write what your teacher tells you

2.8 Step 1 - Identification of requirements

A complete investigation of the existing system is carried out. The objectives, benefits and efficiency are worked out with the staff of the organization and the development team. The **system analyst** studies the existing system extensively and identifies new requirements; the developer collects information to fulfil the needs of the user. Methods of gathering information:

Method	What happens	Example (a library)
1. Observation	At the initial stages the existing system is observed as it is, to gather information. This is important towards making assumptions.	Observation of issuing of books in a library
2. Interviews	An interview gives an opportunity to discuss matters related to personal attitudes openly, so it is useful to identify personal requirements.	Meeting the librarian for a discussion
3. Questionnaires	A questionnaire is used in user requirement gathering. The answers collected are analysed to understand the needs.	Handing over a questionnaire to the librarian for his or her responses
4. Document sample collection	Reports or files kept in the organization under study are examined for information.	Reports and library cards in a library
5. Prototyping	A model (prototype) of the proposed system is developed and shown to the staff and the users for feedback and comments.	A display of screenshots, records and samples

2.9 Step 2 - Designing the solution

At this stage the actual design of the proposed system takes place. Some activities of the design process:

1. Identification of software and software architecture
2. User interface - what the user sees on the screen - and preparation for data storage
3. Identification of the main hardware systems and their components
4. Identification of the dependency of each sub-system
5. Deciding on the required hardware and software to run the system
6. Designing infrastructure for software, databases and user interfaces
7. Planning of tests

"Listen to the user first. The best systems begin with the right questions."

- THE KING OF ICT

2.10 Step 3 - Coding of the solution

The main objective of this stage is coding the proposed system using a suitable computer language. Simple and efficient code is easy to develop and understand. A well-written code is easy to read and understand, and it reduces the **maintenance cost**.

DO IT - PRACTICAL 2 (TEXTBOOK PAGES 57 - 58)

Gather the requirements of a library system. Work in groups: your school wants a computer based library system, and you are the development team.

1	**Observation:** watch how the books are issued in the school library.
2	**Interview:** meet the librarian for a discussion. What does the librarian need?
3	**Questionnaire** and **documents:** give the librarian five questions; examine the library cards.
4	Write the requirements you found in the table below and tell the class.

Method	The requirements we found
Observation	
Interview	
Questionnaire	
Documents	

2.11 Step 4 - Testing and debugging

The main objective of this phase is to **resolve errors**: errors in coding, planning and requirements are corrected. There are several methods to test a system. They are done in this order:

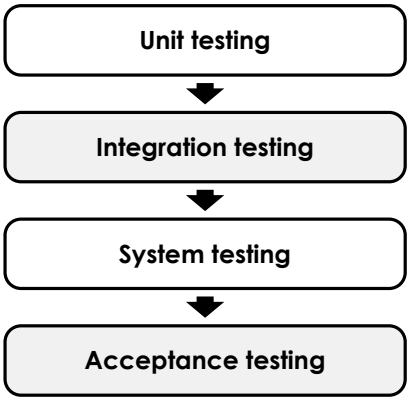


Figure 2.5 - system testing

Testing	What is tested
Unit testing	Each unit is tested: its outputs against the given output. E.g. the units of the Accounts branch and the Establishment branch are tested independently.
Integration testing	Units already tested and debugged are integrated to form the complete system, and the integrated system is tested.
System testing	The complete system is tested for its outputs for the given inputs.
Acceptance testing	At the late stages, with the users: the user uses the system and approves it or requests more improvements.

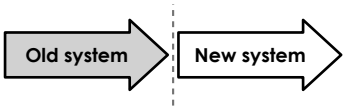
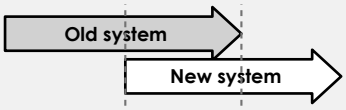
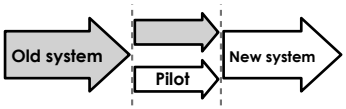
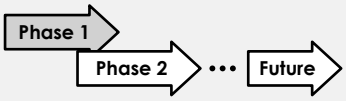
"A plan on paper saves days of work on the computer."

- THE KING OF ICT

WRITING SPACE 4 - The four levels of testing write what your teacher tells you

2.12 Step 5 - Deployment of the system

Deployment means to install and use the system that has been tested and debugged. There are different deployment approaches:

Picture	Approach - what happens	Example (pages 60 - 61)
	**Direct deployment** The existing system is totally terminated and the new system is installed.	The manual system in the library is abandoned and new library management software is installed.
	**Parallel deployment** The existing and the new systems are used in parallel for a certain period. If the new system proves successful, the old system is terminated.	The newly deployed library system is used together with the existing manual system.
	**Pilot deployment** The newly developed system is used first in a selected area.	A system for the whole country is first installed in a selected district.
	**Phased deployment** The new system is deployed in stages. Success with each stage leads to the next, until the old system is completely replaced.	School library books: dictionaries first, course books second and fiction last.

WRITING SPACE 5 - The four deployment approaches write what your teacher tells you

2.13 Step 6 - Maintenance of the system

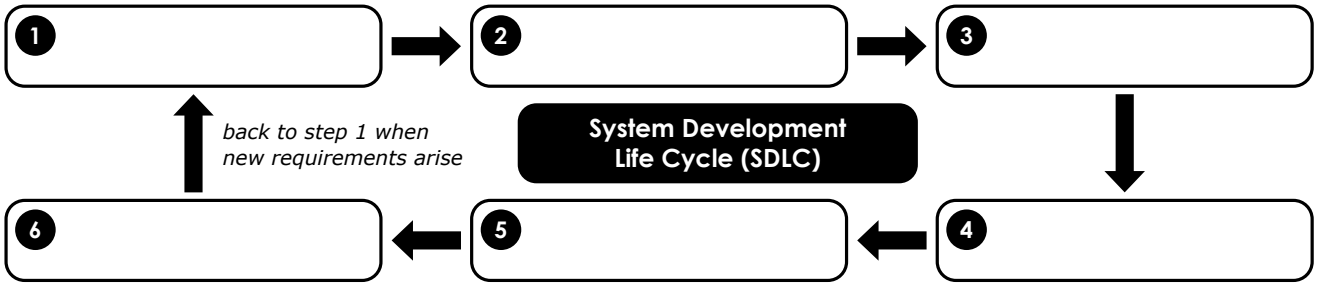
A system that has been installed and in use for a certain period needs to be **maintained** well. The computers, the software and the computer networks are all maintained. With system maintenance, these may be required:

- Developing the system to suit new user requirements
- Issues that surfaced with the use of the system which were not identified during system development
- Adopting new technology into the system to improve efficiency

A Label the life cycle

6 marks

Write the six steps of the SDLC for the numbered boxes.



1. ..... 2. .....

3. ..... 4. .....

5. ..... 6. .....

B Which method?

8 marks

Write the method of gathering information (2.8).

What happens	Method
The developer watches how the books are issued in the library.	
The developer meets the librarian for a discussion.	
The librarian writes answers to a set of questions.	
The reports and the library cards of the library are examined.	
Users comment on screenshots of the proposed system.	
The users talk openly about their personal requirements.	
At the initial stage, the existing system is observed as it is.	
The answers collected are analysed to understand the needs.	

C True or false

6 marks

Write T (true) or F (false) in the box.

1. ☐ The first step of the SDLC is designing the solution.

2. ☐ An interview is useful to identify personal requirements.

3. ☐ The tests are planned in the coding step.

4. ☐ Unit testing is done after acceptance testing.

5. ☐ In direct deployment the existing system is totally terminated.

6. ☐ Computers, software and computer networks are all maintained.

"Simple code is strong code: easy to read, easy to keep."

- THE KING OF ICT

D Which level of testing?

8 marks

Tick one box for each line (2.11).

Which testing?	Unit	Integration	System	Acceptance
Each unit in the system is tested on its own.	☐	☐	☐	☐
Units already tested and debugged are joined and tested together.	☐	☐	☐	☐
The complete system is tested for its outputs for the given inputs.	☐	☐	☐	☐
The user uses the system and approves it or asks for more improvements.	☐	☐	☐	☐
The units of the Accounts branch and the Establishment branch are tested independently.	☐	☐	☐	☐
It is carried out at the late stages of testing, with the users.	☐	☐	☐	☐
The integrated system is tested.	☐	☐	☐	☐
After integration testing, the whole system is tested with given inputs.	☐	☐	☐	☐

E Which deployment approach?

8 marks

Tick one box for each line (2.12).

Which approach?	Direct	Parallel	Pilot	Phased
The manual library system is abandoned and new library software is installed.	☐	☐	☐	☐
The new library system is used together with the existing manual system.	☐	☐	☐	☐
A system for the whole country is first installed in a selected district.	☐	☐	☐	☐
Dictionaries first, course books second and fiction last.	☐	☐	☐	☐
The existing system is totally terminated and the new system is installed.	☐	☐	☐	☐
A revenue collection system for vehicles is first installed in one province.	☐	☐	☐	☐
The old and the new systems are used together for a certain period.	☐	☐	☐	☐
The new system is deployed in stages: success with each stage leads to the next.	☐	☐	☐	☐

"Every error you find before the users do is a gift to them."

- THE KING OF ICT

CHAPTER 3

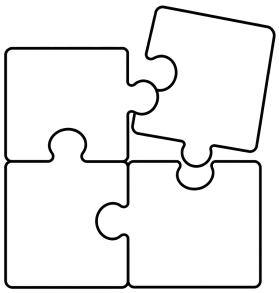
Life Cycle Models

? ALL AT ONCE OR PIECE BY PIECE?

"Shall we build the whole library system and show it to the librarian at the end?"

"Or build one small part at a time, and let the system grow in every round?"

Each **SDLC model** develops a system in a different way. Meet two of them.



2.14 System development life cycle models

There are several models of the System Development Life Cycle. Each model facilitates the system development in a different way. Some example models are:

1 Waterfall Model	2 Iterative Incremental Model	3 Prototyping Model	4 Spiral Model

2.15 The waterfall model

The waterfall model consists of several stages of system development. It shows **linear development** with the life cycle. To develop a system with the waterfall model, consider these:

- First identify the requirements.
- Complete one stage before going to the next phase.
- The result of the developed system is found at the final stage.

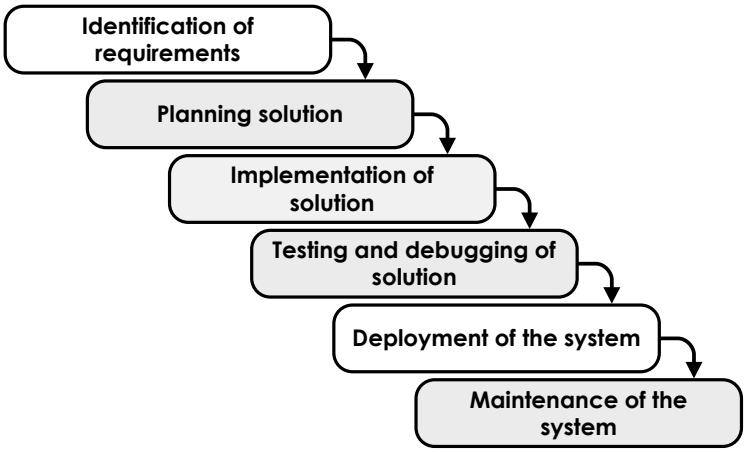


Figure 2.6 - the waterfall model

! ITS LIMITATION

The system developed may be different from what the user wants, as the user sees the system only at the end of the life cycle.

"Big systems are built one small step at a time."

- THE KING OF ICT

WRITING SPACE 6 - The waterfall model write what your teacher tells you

2.16 The iterative incremental model

The iterative incremental model is also used as a system development model. It begins with initial planning, goes round the cycle again and again, and ends with deployment:

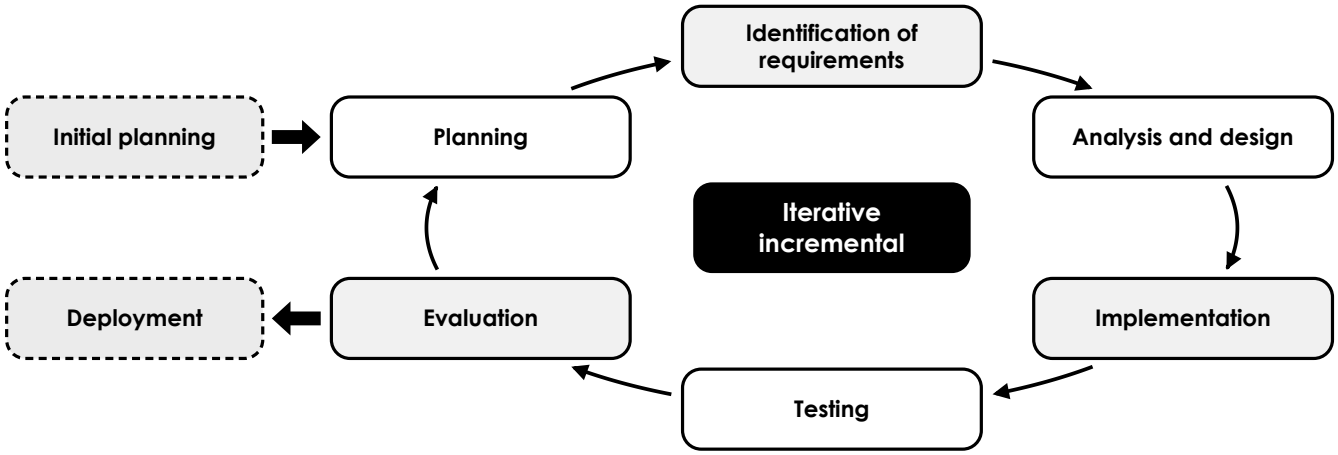


Figure 2.7 - the iterative incremental life cycle

Some features of the iterative incremental model:

- The main objective is to develop one small component at a time, in an iterative fashion. Each iteration develops more components of the system.
- System developers have the advantage of using the knowledge gained in the earlier iterations.
- The first step begins with the rough idea of the system requirement.
- The system gets developed more and more, incrementally, until the last stage.
- At each incremental stage the system becomes larger, with more and more components completed.

WRITING SPACE 7 - The iterative incremental model write what your teacher tells you

DO IT - PRACTICAL 3 (TEXTBOOK ACTIVITY, PAGE 63)

Compare and contrast the two models. Work in pairs.

1	Read 2.15 and 2.16 again.
2	Write how each model develops a system in the table below.
3	Find one limitation of the waterfall model and one advantage of the iterative incremental model.
4	Discuss your answers with the class.

	Waterfall model	Iterative incremental model
How it develops		
When the user sees it		
The first step		

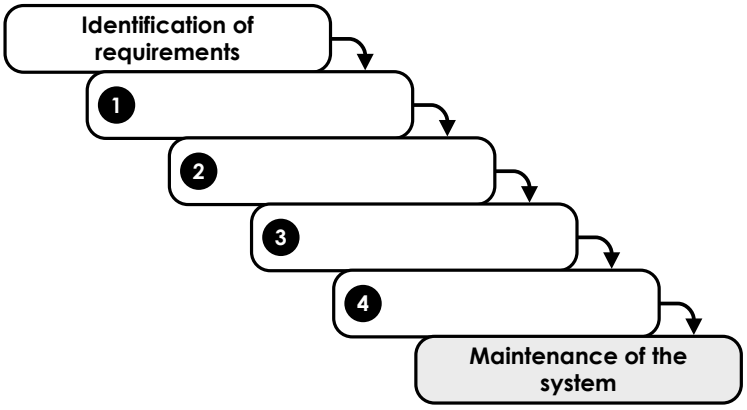
A Complete the waterfall

4 marks

Write the stages of Figure 2.6. Two names in the box are extra.

WORD BANK

Planning solution	Implementation of solution	Testing and debugging of solution
Deployment of the system	Evaluation	Initial planning



1. .....
2. .....
3. .....
4. .....

B True or false

5 marks

Write T (true) or F (false) in the box.

1. ☐ Each SDLC model facilitates system development in a different way.

2. ☐ In the waterfall model, the next stage can start before a stage is complete.

3. ☐ In the waterfall model the user sees the system at the beginning of the life cycle.

4. ☐ In the iterative incremental model each iteration develops more components.

5. ☐ The iterative incremental model starts with a complete, final list of requirements.

C Waterfall or iterative incremental?

8 marks

Which model is it? Tick one box for each line.

The model ...	Waterfall	Iterative incremental
Complete one stage before going to the next phase.	☐	☐
Develop one small component at a time.	☐	☐
The result of the developed system is found at the final stage.	☐	☐
Developers use the knowledge gained in the earlier iterations.	☐	☐
It shows linear development.	☐	☐
The first step begins with a rough idea of the system requirement.	☐	☐
The user sees the system only at the end of the life cycle.	☐	☐
At each stage the system becomes larger, with more components.	☐	☐

D Name the four models

4 marks

Write the four SDLC models named in the textbook (2.14).

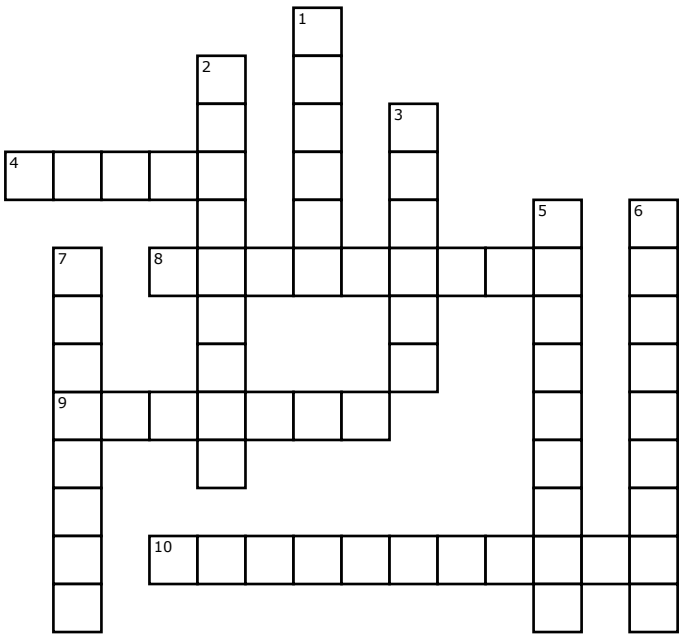
1. ..... 2. .....

3. ..... 4. .....

E Crossword

10 marks

All the words are in this unit. Write one letter in each square.



ACROSS

4. ... deployment: first in a selected area (5)
8. A model of the proposed system shown to users for comments (9)
9. System ... : studies the existing system and finds new requirements (7)
10. Step 6 of the SDLC (11)

DOWN

1. Input is converted to this by processing (6)
2. The model that shows linear development (9)
3. A collection of components that interact to achieve a specific task (6)
5. Testing and ... : resolving the errors (9)
6. Meeting the librarian for a discussion (9)
7. ... deployment: old and new systems used together (8)

"Finish one step well before you run to the next."

- THE KING OF ICT

REVISION

Unit 2 in One Page

Read this the night before the test

Topic	The key facts
System	A collection of components that interact to achieve a specific task. If the components are not interconnected, it does not work.
Basic components	Input, process, output: input is converted to output by processing. School: children - teaching/learning - good citizens.
Information system	Converts data into information. Two kinds: manual and computer based.
Manual system	All processes are done manually: more room for error, less efficient, filing cabinets, open to threats.
Computer based	Minimal errors, efficient, large data in a small space (database software), secure with backups and passwords.
SDLC	Identification of requirements, designing, coding, testing and debugging, deployment, maintenance - back to step 1 when new requirements arise.
Gathering information	Observation, interviews, questionnaires, document sample collection, prototyping.
Designing	The actual design: software architecture, user interface and data storage, hardware, sub-systems, infrastructure, planning of tests.
Coding	The system is coded in a suitable language. Simple, efficient, well-written code is easy to read and reduces the maintenance cost.
Testing	Resolves errors: unit, integration, system and acceptance testing, in this order.
Deployment	Install and use: direct, parallel, pilot (a selected area) or phased (in stages).
Maintenance	New user requirements, issues not found during development, new technology for efficiency.
SDLC models	Waterfall (linear; the user sees the system at the end), iterative incremental (one small component at a time), prototyping, spiral.

★ TICK WHEN YOU CAN DO IT

- ☐ I can explain what a system is and name its three basic components.
- ☐ I can find the input, the process and the output of a system.
- ☐ I can compare manual and computer based information systems.
- ☐ I can name the six steps of the SDLC in order.
- ☐ I can name the five methods of gathering information.
- ☐ I can name the four levels of testing and the four deployment approaches.
- ☐ I can compare the waterfall model and the iterative incremental model.

WORD POWER

Unit 2 Word List

Twenty-six words you must know

Word	What it means
System	Components that interact to achieve a specific task.
Component	One part of a system.
Input	What a system receives; processing converts it to output.
Process	The work that converts the input into the output.
Output	The result that a system gives.
Information system	A system which converts data into information.
Manual system	All its processes are done by people, without computers.
Computer based system	Converts data into information using a computer.
SDLC	System Development Life Cycle: six steps.
Requirements	What the users need from the new system.
System analyst	Studies the existing system; finds new requirements.
Observation	Watching the existing system as it is.
Interview	A discussion that finds personal requirements.
Questionnaire	Questions for users; the answers show their needs.
Prototype	A model of the proposed system, shown for comments.
User interface	What the user sees on the screen.
Debugging	Finding and correcting the errors of a system.
Unit testing	Each unit of the system is tested on its own.
Integration testing	The tested units are joined and tested together.
Acceptance testing	The users approve the system or ask for changes.
Deployment	Installing and using the tested and debugged system.
Pilot deployment	The new system is used first in a selected area.
Phased deployment	The new system is deployed in stages.
Maintenance	Keeping the system working well while it is in use.
Waterfall model	Linear: each stage is completed before the next.
Iterative incremental model	Develops one small component at a time.

"Learn the words today, and you will speak like a system analyst tomorrow."

- THE KING OF ICT

"Read every question twice. Answer once, clearly."

- THE KING OF ICT

END OF UNIT

Unit 2 Test

Time: 45 minutes Total: 40 marks

Name _____ **Class** _____ **Marks** ____

1 Choose the correct answer

8 marks

Underline your answer.

1. The output of the school system (Example 1) is
   (a) teachers (b) school books (c) good citizens (d) classrooms
2. An information system converts data into
   (a) input (b) information (c) files (d) cabinets
3. In a manual system all the processes are done
   (a) by computer (b) online (c) by robots (d) manually
4. Who studies the existing system extensively?
   (a) librarian (b) principal (c) system analyst (d) users
5. The user interface is what the user ... on the screen.
   (a) sees (b) hears (c) prints (d) deletes
6. Which is a benefit of a computer based library system?
   (a) paper registers (b) e-books for members
   (c) longer queues (d) lost books
7. Which SDLC model shows linear development?
   (a) spiral (b) prototyping (c) iterative (d) waterfall
8. A - The school is an example of a system. B - A weather forecasting system outputs tomorrow's weather. C - The user interface is what the user hears. Which are correct?
   (a) A only (b) A and B (c) B and C (d) A, B and C

2 Fill in the blanks

8 marks

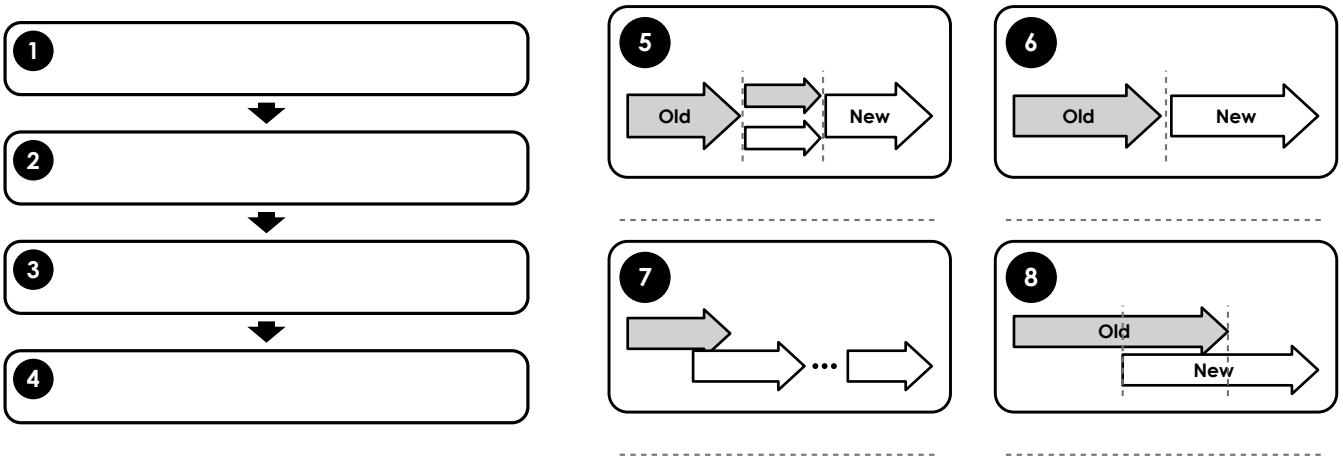
One word in each blank.

1. In the school system (Example 1), the input is the ..... .
2. A computer based library system calculates ..... for late returns.
3. In the manual system, the student's file is found by the ..... number.
4. In the manual system the principal asks for a progress ..... .
5. Humidity is one of the ..... of a weather forecasting system.
6. A library system uses a ..... code for issues and receipts.
7. An online library system allows remote ..... and searching for books.
8. The connected machine of Figure 2.2 works as a ..... .

3 Label and name the pictures

8 marks

Boxes 1 - 4: write the four levels of testing in order. Cards 5 - 8: name the deployment.



4 Answer these

16 marks

Write in the space given.

1. What is a system? (2 marks)

2. Write two differences between manual systems and computer based information systems. (2 marks)

3. Name the six steps of the System Development Life Cycle in order. (3 marks)

4. Name three methods of gathering information for the identification of requirements. (3 marks)

5. Why does a system need maintenance? Write two reasons. (2 marks)

6. Write the limitation of the waterfall model. (2 marks)

7. Write two features of the iterative incremental model. (2 marks)